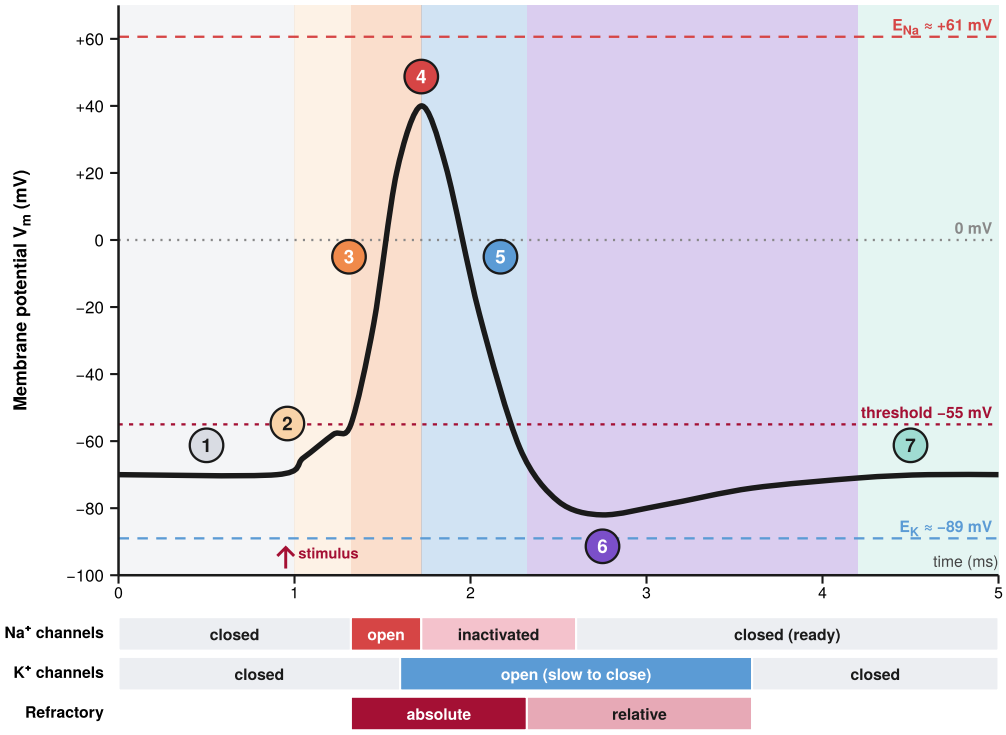
Harvard

Fundamentals of Neuroscience · MicroBachelors® Program

The Action Potential — Phases & Ion Channels

MCB80x · voltage-gated Na^+ and K^+ channels · refractory periods

ACTION POTENTIAL — V_m vs. TIME



Typical mammalian neuron at 37 °C. E values from your formula sheet's Example A.

PHASES

#	Phase	What happens · exam fact
1	**Resting potential** ≈ −70 mV	K^+ leak channels dominate, so V_m sits near E_K. **FACT** Rest is set mainly by K^+ permeability (Goldman equation).
2	**Depolarization** to threshold ≈ −55 mV	Stimulus or summed EPSPs push V_m upward. **FACT** Below threshold → no spike. All-or-none.
3	**Rising phase** Na^+ in	Voltage-gated Na^+ channels open; Na^+ rushes in, opening more channels (positive feedback). **FACT** Tetrodotoxin (TTX) blocks these Na^+ channels.
4	**Peak / overshoot** ≈ +40 mV	Na^+ channels inactivate; V_m heads toward but never reaches E_{Na}. **FACT** Inactivation gate = "ball-and-chain" plug.
5	**Falling phase** K^+ out	Delayed voltage-gated K^+ channels open; K^+ flows out and repolarizes. **FACT** TEA blocks K^+ channels → spike stays prolonged.
6	**Undershoot** ≈ −80 mV	K^+ channels close slowly, so V_m dips toward E_K. **FACT** Also called after-hyperpolarization.
7	**Return to rest** ≈ −70 mV	K^+ channels close; leak channels restore resting potential. **FACT** Na^+/K^+ pump keeps gradients long-term; not needed per spike.

Refractory periods

- **Absolute:** no new spike at any strength — Na^+ channels are open or inactivated.
- **Relative:** a stronger stimulus can fire — K^+ still open, some Na^+ recovered.
- Refractoriness makes spikes travel **one way** and caps firing rate.

Personal MCB80x study chart — not an official HarvardX/edX publication.